Optimization

Francesca Maggioni

Department of Management, Information and Production Engineering
University of Bergamo

Lesson 4

EXERCISE

Consider the following training set in $\mathbb{R}^2$:

Point	Label
$x^1 = (1, 0)$	$y_1 = +1$
$x^2 = (0, 1)$	$y_2 = +1$
$x^3 = (-1, 0)$	$y_3 = -1$
$x^4 = (0, -1)$	$y_4 = -1$

We consider the hard-margin Support Vector Machine problem:

$$\begin{aligned} \min_{w,\theta} \quad & \frac{1}{2}\|w\|^2 \\ \text{s.t.} \quad & y_p(w^\top x^p + \theta) \geq 1, \quad p = 1, \dots, 4. \end{aligned}$$

UNIVERSITÀ DEGLI STUDI DI BERGAMO

Answer the following questions:

1. Write the problem in standard inequality form $\leq$.
2. Construct the Lagrangian function.
3. Write the Karush–Kuhn–Tucker (KKT) conditions.
4. Derive the Wolfe dual problem.
5. Solve the dual problem.
6. Determine the optimal w and θ.
7. Identify the support vectors.
8. Compute the geometric margin of the classifier.

UNIVERSITÀ DEGLI STUDI DI BERGAMO

Solution

1. Standard form

The constraints

$$y_p(w^\top x^p + \theta) \geq 1$$

can be rewritten as

$$1 - y_p(w^\top x^p + \theta) \leq 0.$$

Define

$$g_p(w, \theta) = 1 - y_p(w^\top x^p + \theta).$$

2. Lagrangian

$$\mathcal{L}(w, \theta, \alpha) = \frac{1}{2} w^\top w + \sum_{p=1}^{4} \alpha_p \left(1 - y_p(w^\top x^p + \theta)\right), \quad \alpha_p \geq 0.$$

Solution

3. KKT conditions

Stationarity:

$$\nabla_w \mathcal{L} = w - \sum_{p=1}^{4} \alpha_p y_p x^p = 0$$

$$\Rightarrow w = \sum_{p=1}^{4} \alpha_p y_p x^p$$

$$\frac{\partial \mathcal{L}}{\partial \theta} = -\sum_{p=1}^{4} \alpha_p y_p = 0$$

$$\Rightarrow \sum_{p=1}^{4} \alpha_p y_p = 0$$

Primal feasibility:

$$y_p(w^\top x^p + \theta) - 1 \geq 0$$

UNIVERSITÀ DEGLI STUDI DI BERGAMO

Solution

Dual feasibility:

$$\alpha_p \geq 0$$

Complementary slackness:

$$\alpha_p \left[y_p(w^\top x^p + \theta) - 1 \right] = 0$$

UNIVERSITÀ DEGLI STUDI DI BERGAMO

Solution

4. Wolfe dual

Substituting $w = \sum \alpha_p y_p x^p$ into the Lagrangian gives:

$$\max_{\alpha} \sum_{p=1}^{4} \alpha_p - \frac{1}{2} \sum_{p,q=1}^{4} \alpha_p \alpha_q y_p y_q (x^p)^\top x^q$$

subject to

$$\alpha_p \geq 0, \qquad \sum_{p=1}^{4} \alpha_p y_p = 0.$$

5. Solving the dual

By symmetry, the optimal solution satisfies:

$$\alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = \frac{1}{2}.$$

UNIVERSITÀ DEGLI STUDI DI BERGAMO

Solution

6. Optimal w

$$w = \sum_{p=1}^{4} \alpha_p y_p x^p = \frac{1}{2}\left[(1,0) + (0,1) + (1,0) + (0,1)\right] = (1,1).$$

Using any support vector:

$$y_1(w^\top x^1 + \theta) = 1 \Rightarrow (1)(1+\theta) = 1 \Rightarrow \theta = 0.$$

7. Support vectors

All four points satisfy the margin equality, so all are support vectors.

8. Geometric margin

$$\gamma = \frac{1}{\|w\|} = \frac{1}{\sqrt{2}}.$$

UNIVERSITÀ DEGLI STUDI DI BERGAMO

Slater's Conditions for Strong Duality

Are there any conditions, beyond convexity, that guarantee strong duality?

Among these conditions, called *constraints qualifications*, there are the **Slater's condition** that guarantee strong duality to hold if f and $g_1 \dots g_m$ are convex.

Let $\mathcal{D}$ be the set on which f and each g_i are defined, that is $\mathcal{D} := dom(f) \cap \mathcal{G}$, where $\mathcal{G} := \bigcap_{i=1}^{m} dom(g_i)$. We say that $g_1, \dots, g_m$ satisfy the **Slater's condition** if

$$g_i(\bar{\mathbf{x}}) < 0, \quad i = 1, \dots, m,$$

for some $\bar{\mathbf{x}}$ in the *relative interior* of $\mathcal{D}$. This is equivalent to state that $\bar{\mathbf{x}}$ is *strictly feasible*, i.e. all m constraints are satisfied with strict inequality.

Theorem (Slater's condition)

If f and $g_1 \dots g_m$ are convex and $g_1 \dots g_m$ satisfy the **Slater's condition**, then **strong duality holds**.

UNIVERSITÀ DEGLI STUDI DI BERGAMO

Slater's Conditions for Strong Duality

The Slater's condition for strong duality can be **refined** if some of the constraint functions $g_1, \ldots, g_m$ are *affine*.

Let $g_1, \ldots, g_k$ be **the first k affine functions**, with $k \leq m$. We say that the convex functions $g_1 \ldots g_m$ satisfy the **refined Slater's condition** if

$$g_i(\bar{\mathbf{x}}) \leq 0, \quad i = 1, \ldots, k \qquad g_i(\bar{\mathbf{x}}) < 0 \quad i = k+1, \ldots, m,$$

for some $\bar{\mathbf{x}}$ in the *relative interior* of $\mathcal{D}$. This condition is equivalent to state that $\bar{\mathbf{x}}$ is *strictly feasible* **only for the nonaffine constraints**, while the affine inequalities do not need to hold with strict inequality.

Remark

Slater's and refined Slater's conditions not only **guarantee strong duality** when f and $g_1 \ldots g_m$ are convex. They also imply that the **dual optimal value** is **attained** when the *dual problem* is bounded from below.

UNIVERSITÀ DEGLI STUDI DI BERGAMO

Example: Least-squares solution of linear equations

We consider the problem

$$\begin{aligned} \min \quad & \mathbf{x}^\top \mathbf{x} \\ \text{s.t.} \quad & A\mathbf{x} = \mathbf{b}, \end{aligned}$$

where $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{b} \in \mathbb{R}^p$, and $A \in \mathbb{R}^{p \times n}$. This problem has **no inequality** constraints and p **linear equality** constraints.

Its Lagrangian function is

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = \mathbf{x}^\top \mathbf{x} + \boldsymbol{\lambda}^\top (A\mathbf{x} - \mathbf{b}),$$

which is a **convex quadratic** function of $\mathbf{x}$, with $\boldsymbol{\lambda} \in \mathbb{R}^p$.

As mentioned before for the WDP, the **dual function** of $\mathcal{L}(\mathbf{x}, \lambda)$ is

$$\theta(\boldsymbol{\lambda}) = \inf_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = \inf_{\mathbf{x}} \left(\mathbf{x}^\top \mathbf{x} + \boldsymbol{\lambda}^\top (A\mathbf{x} - \mathbf{b}) \right).$$

Slater's Conditions for Strong Duality

The **convexity** of $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda})$ implies that the minimizing $\mathbf{x}^*$ can be found from the **optimality condition** $(*)$, that is

$$\nabla \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = 2\mathbf{x} + A^\top \boldsymbol{\lambda} = \mathbf{0},$$

which yields to $\mathbf{x}^* = -\frac{1}{2} A^\top \boldsymbol{\lambda}$.

Then, the dual function can be written as

$$\begin{aligned}\theta(\boldsymbol{\lambda}) =& \mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}) = \left(-\frac{1}{2} A^\top \boldsymbol{\lambda} \right)^\top \left(-\frac{1}{2} A^\top \boldsymbol{\lambda} \right) + \boldsymbol{\lambda}^\top \left(A\Big(-\frac{1}{2} A^\top \boldsymbol{\lambda} \Big) - \mathbf{b} \right) = \\ &= -\frac{1}{4} \boldsymbol{\lambda}^\top A A^\top \boldsymbol{\lambda} - \mathbf{b}^\top \boldsymbol{\lambda}.\end{aligned}$$

which is a **concave quadratic** function for which **weak duality** holds.

Therefore, $\theta(\boldsymbol{\lambda})$ provides a **lower bound** for the primal problem, that is

$$-\frac{1}{4} \boldsymbol{\lambda}^\top A A^\top \boldsymbol{\lambda} - \mathbf{b}^\top \boldsymbol{\lambda} \leq \inf\{\mathbf{x}^\top \mathbf{x} | A\mathbf{x} = \mathbf{b}\}.$$

What about the **strong duality**?

UNIVERSITÀ DEGLI STUDI DI BERGAMO

Slater's Conditions for Strong Duality

Starting from the **dual problem**, formulated as

$$\max_{\boldsymbol{\lambda}} \theta(\boldsymbol{\lambda}) = \max_{\boldsymbol{\lambda}} \Big(-\frac{1}{4}\boldsymbol{\lambda}^\top A A^\top \boldsymbol{\lambda} - \mathbf{b}^\top \boldsymbol{\lambda} \Big),$$

we note it is an **unconstrained concave maximization problem**.

This guarantees that the Slater's condition is always satisfied if the **primal problem is feasible**.

Therefore, the **strong duality** always holds if $\mathbf{b} \in \langle A_1, A_2, \dots A_n \rangle$, where A_i denotes the i-th column of A.

Remark

Strong duality holds even when $\mathbf{b} \notin \langle A_1, A_2, \dots A_n \rangle$, that is there exists $\mathbf{z} \in \mathbb{R}^p$ such that

$$A^\top \mathbf{z} = \mathbf{0} \text{ and } \mathbf{b}^\top \mathbf{z} \neq \mathbf{0}.$$

It follows, indeed, that $\theta(\boldsymbol{\lambda})$ is also unbounded above along $\{t\mathbf{z} \,|\, t \in \mathbb{R}\}$.

UNIVERSITÀ DEGLI STUDI DI BERGAMO

Linear Non-Separable SVM Model

When the **training set** is **not linearly separable** the problem

$$\begin{aligned} \min \quad & \frac{1}{2}||w||^2 \\ s.t. \quad & y_p(w^\top x^p + \theta) \geq 1 \quad p = 1, \dots, P. \end{aligned}$$

has **empty feasible region**. In this case **slack variables** ξ_p must be introduced and the new optimization problem is:

Soft Margin SVM

$$\begin{aligned} \min \quad & \frac{1}{2}||w||^2 + C\sum_{p=1}^{P} \xi_p \\ s.t. \quad & y_p(w^\top x^p + \theta) \geq 1 - \xi_p \quad p = 1, \dots, P \\ & \xi_p \geq 0 \quad p = 1, \dots, P. \end{aligned} \tag{**}$$

UNIVERSITÀ DEGLI STUDI DI BERGAMO

Linear Non-Separable SVM Model

- For the hyperplane $H_{w,\theta}$ **if a point x^p is not correctly classified then the corresponding quantity ξ_p is bigger than 1.** In fact:

$$\text{if } y_p = +1 \text{ and } w^\top x^p - \theta \leq 0 \text{ then } \xi_p = 1 - w^\top x^p + \theta \geq 1,$$

$$\text{if } y_p = -1 \text{ and } w^\top x^p + \theta \geq 0 \text{ then } \xi_p = 1 + w^\top x^p + \theta \geq 1.$$

- Moreover, values of ξ_p in the interval $(0, 1)$ correspond to points correctly classified by the hyperplane but inside the separation zone.
- The objective function includes two terms:
  1. the first term corresponds to margin maximization: $\frac{1}{2}||w||^2$;
  2. the second term is an upper bound on the number of incorrectly classified training set points: $\sum \xi_p$.

UNIVERSITÀ DEGLI STUDI DI BERGAMO

Optimality Conditions and Dual Problem

The **Lagrangian function** associated to problem (**) is:

$$\mathcal{L}(w, \theta, \xi, \alpha) = \frac{1}{2} w^\top w + C \sum_{p=1}^{P} \xi_p - \sum_{p=1}^{P} \alpha_p \left(y_p (w^\top x^p + \theta) - 1 + \xi_p \right)$$

and the Karush–Kuhn–Tucker optimality conditions are:

$$w = \sum_{p=1}^{P} \alpha_p y_p x^p$$

$$\sum_{p=1}^{P} \alpha_p y_p = 0$$

$$0 \leq \alpha_p \leq C \qquad p = 1, \ldots, P$$

$$\xi_p \geq 0 \qquad p = 1, \ldots, P$$

$$\xi_p (C - \alpha_p) = 0 \qquad p = 1, \ldots, P$$

$$y_p (w^\top x^p + \theta) - 1 + \xi_p \geq 0 \qquad p = 1, \ldots, P$$

$$\alpha_p \left[y_p (w^\top x^p + \theta) - 1 + \xi_p \right] = 0 \qquad p = 1, \ldots, P.$$

UNIVERSITÀ DEGLI STUDI DI BERGAMO

Optimality Conditions and Dual Problem

The **dual problem** is now:

$$\begin{aligned} \min_{\alpha} \quad & \frac{1}{2}\left(\sum_{p=1}^{P}\alpha_p y_p x^p\right)^\top \left(\sum_{q=1}^{P}\alpha_q y_q x^q\right) - \sum_{p=1}^{P}\alpha_p \\ s.t. \quad & \sum_{p=1}^{P}\alpha_p y_p = 0 \\ & 0 \le \alpha_p \le C \qquad\qquad p = 1, \dots, P \end{aligned}$$

or, in more compact form:

$$\begin{aligned} \min_{\alpha} \quad & \tfrac{1}{2}\alpha^\top Q\alpha - e^\top \alpha \\ s.t. \quad & y^\top \alpha = 0 \\ & 0 \le \alpha \le Ce \end{aligned}$$

where e is a vector with all components equal to 1, $Q \in \mathbb{R}^{P\times P}$ and:

$$Q_{pq} = y_p y_q K(x^p, x^q).$$

UNIVERSITÀ DEGLI STUDI DI BERGAMO

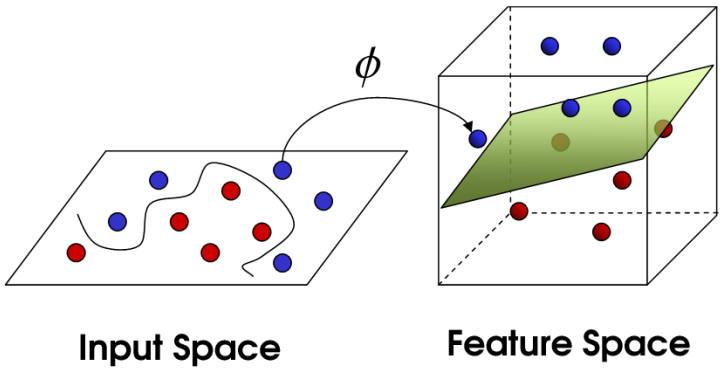
Remark

- *Here the support vectors are either free support vectors corresponding to* $0 < \alpha_p^* < C$ *or bounded support vectors, corresponding to* $\alpha_p^* = C$*.*
- *Again, the vector* w^* *is a linear combination of support vectors while the scalar* θ^* *can be calculated using any free support vector.*

UNIVERSITÀ DEGLI STUDI DI BERGAMO

Non-Linear SVM Model: Learning in the Feature Space

- The idea behind non-linear SVM is to map the input space into a **feature space** where they are linearly separable.
- It must be recalled that in linear SVM models the dual problem can be easily constructed using the function $K(\cdot, \cdot)$. Moreover, the decision function (**??**) only utilizes the function $K(\cdot, \cdot)$.
- Let $\mathcal{F}$ be an Hilbert space (the feature space) whose dimension is bigger than n and possibly infinite, and let $\phi : \mathbb{R}^n \mapsto \mathcal{F}$ be a mapping from $\mathbb{R}^n$ to the feature space.

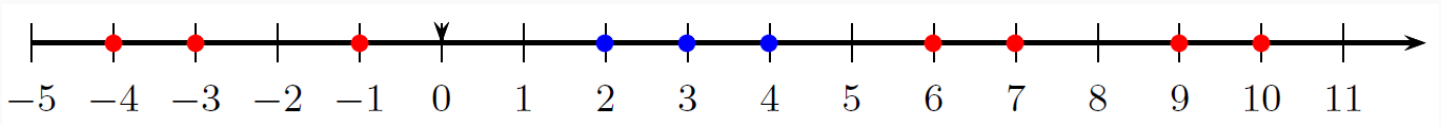


Example

As an example, consider the following 1-dimesional training set with $P = 10$:

p	1	2	3	4	5	6	7	8	9	10
x_p	-4	-3	-1	2	3	4	6	7	9	10
y_p	-1	-1	-1	$+1$	$+1$	$+1$	-1	-1	-1	-1

represented in the figure below:



This training set is not linearly separable. A possible classification function would be $h(x) = \text{sign}\big(x(x-5)\big)$.

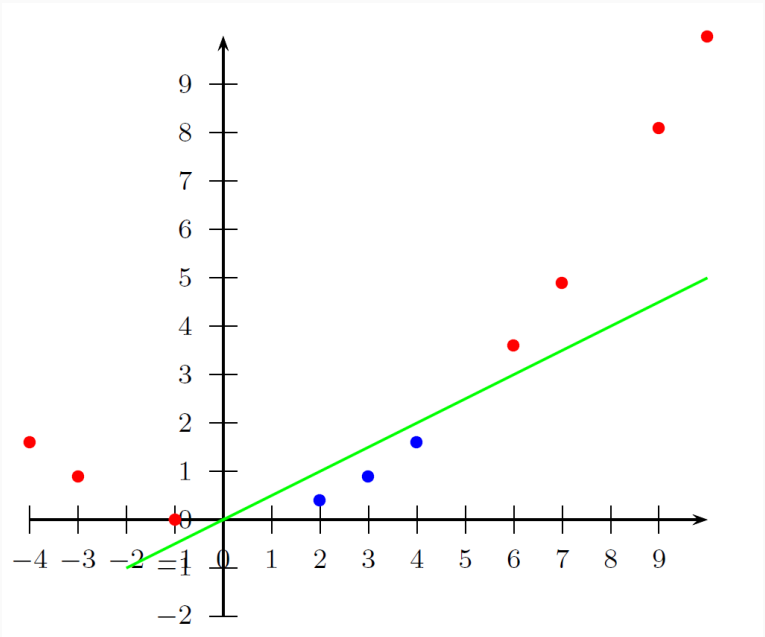
UNIVERSITÀ DEGLI STUDI DI BERGAMO

Example

Now, consider the map $\phi : \mathbb{R} \to \mathbb{R}^2$ given by:

$$\phi(x) := \begin{bmatrix} x \\ x^2 \end{bmatrix} \in \mathbb{R}^2.$$

In $\mathbb{R}^2$ the mapped values are linearly separable as clearly showed in the figure:



UNIVERSITÀ DEGLI STUDI DI BERGAMO

Non-Linear SVM Model: Learning in the Feature Space

For the non-linear SVM model the optimization problem is now:

Non Linear SVM

$$\begin{aligned} \min_{w,\theta,\xi} \quad & \frac{1}{2}||w||^2 + C\sum_{p=1}^{P}\xi_p & & \\ s.t. \quad & y_p(w^\top \phi(x^p) + \theta) \geq 1 - \xi_p & p = 1,\dots,P & \quad (***) \\ & \xi_p \geq 0 & p = 1,\dots,P. & \end{aligned}$$

UNIVERSITÀ DEGLI STUDI DI BERGAMO

For problem (***) the Karush–Kuhn–Tucker optimality conditions are:

KKT Conditions Non Linear SVM

$$w = \sum_{p=1}^{P} \alpha_p y_p \phi(x^p)$$

$$\sum_{p=1}^{P} \alpha_p y_p = 0$$

$$0 \leq \alpha_p \leq C \qquad p = 1, \ldots, P$$

$$\xi_p \geq 0 \qquad p = 1, \ldots, P$$

$$\xi_p (C - \alpha_p) = 0 \qquad p = 1, \ldots, P$$

$$y_p (w^\top \phi(x^p) + \theta) - 1 + \xi_p \geq 0 \qquad p = 1, \ldots, P$$

$$\alpha_p \left[y_p (w^\top \phi(x^p) + \theta) - 1 + \xi_p \right] = 0 \qquad p = 1, \ldots, P.$$

UNIVERSITÀ DEGLI STUDI DI BERGAMO

The **dual problem** is now:

$$\begin{aligned} \min_{\alpha} \quad & \tfrac{1}{2}\alpha^\top Q\alpha - e^\top \alpha \\ s.t. \quad & y^\top \alpha = 0 \\ & 0 \leq \alpha \leq Ce \end{aligned}$$

where e is a vector with all components equal to 1, $Q \in \mathbb{R}^{P\times P}$ and:

$$Q_{pq} = y_p y_q K(x^p, x^q) \quad \text{with} \quad K(x^p, x^q) = \phi(x^p)^\top \phi(x^q).$$

Remark

The function $K : \mathbb{R}^N \times \mathbb{R}^N \to \mathbb{R}$ is called the **Kernel function**.

UNIVERSITÀ DEGLI STUDI DI BERGAMO

Once again, when the dual problem has been solved (let α^* be its optimal solution) the optimal hyperplane H_{w^*,θ^*} can be obtained by:

- $w^* = \sum_{p=1}^{P} \alpha_p^* y_p \phi(x^p)$ (see first KKT condition);
- if p' is a support vector (that is, $0 < \alpha_{p'}^* < C$), then:

$$y_{p'}(w^\top \phi(x^p) + \theta^*) - 1 = y_{p'} \left(\sum_{p=1}^{P} \alpha_p^* y_p K(x^p, x^{p'}) + \theta^* \right) - 1 = 0$$

from which θ^* can be easily computed.

Then, the decision function $h(\cdot)$ for non-linear SVM is:

$$h(x) = \text{sign}\left({w^*}^\top \phi(x) + \theta^* \right) = \text{sign}\left(\sum_{p=1}^{P} \alpha_p^* y_p K(x^p, x) + \theta^* \right). \qquad (\diamond\diamond)$$

Kernels

Remark

- *The difficulty in an explicit use of the mapping $\phi(\cdot)$ to construct the* **Feature Space** *is that the resulting space can be* **extremely high-dimensional**.
- *However, it must be noticed that both in calculating θ^* and in evaluating the decision function on a new point, the actual form of the function $\phi(\cdot)$ is not necessary but it is only necessary to calculate the Kernel function $K(\cdot, \cdot)$.*

UNIVERSITÀ DEGLI STUDI DI BERGAMO

Example

Consider, for example the following function:

$$\phi : x \in \mathbb{R}^2 \mapsto \begin{bmatrix} x_1^2 \\ \sqrt{2}x_1x_2 \\ x_2^2 \end{bmatrix}.$$

Then:

$$\phi(\bar{x})^\top \phi(\hat{x}) = \begin{bmatrix} \bar{x}_1^2 & \sqrt{2}\bar{x}_1\bar{x}_2 & \bar{x}_2^2 \end{bmatrix} \begin{bmatrix} \hat{x}_1^2 \\ \sqrt{2}\hat{x}_1\hat{x}_2 \\ \hat{x}_2^2 \end{bmatrix} = \left(\bar{x}^\top \hat{x}\right)^2$$

and

$$K(\bar{x}, \hat{x}) = \left(\bar{x}^\top \hat{x}\right)^2.$$

UNIVERSITÀ DEGLI STUDI DI BERGAMO

Kernels

Therefore, a fundamental question is under which hypothesis a function

$$K : \mathbb{R}^N \times \mathbb{R}^N \to \mathbb{R}$$

is a Kernel function, that is

$$K(\bar{x}, \hat{x}) = \phi(\bar{x})^\top \phi(\hat{x})$$

for some function $\phi : \mathbb{R}^N \to \mathcal{F}$.

A related question is for which mappings $\phi(\cdot)$ is easy to evaluate the scalar product $\phi(\bar{x})^\top \phi(\hat{x})$?

Clearly, due to the properties of the inner product, the matrix $K \in \mathbb{R}^{P \times P}$ with

$$K_{pq} = K(x^p, x^q)$$

must be **symmetric**.

UNIVERSITÀ DEGLI STUDI DI BERGAMO

The answer is provided by the **Mercer Theorem**.

Theorem

Let $\mathcal{X}$ be a measure space and let $L^2(\mathcal{X})$ be the Hilbert space of square-integrable functions $f : \mathcal{X} \to \mathbb{R}$. Let $K \in L^2(\mathcal{X} \times \mathcal{X})$. K is a kernel if and only if there exists some feature map $\phi : \mathcal{X} \to \mathcal{H}$ where $\mathcal{H}$ is a Hilbert space such that

$$K(\bar{x}, \hat{x}) = \phi(\bar{x})^\top \phi(\hat{x}).$$

A symmetric $K \in L^2(\mathcal{X} \times \mathcal{X})$ is a kernel if and only if:

$$\forall f \in L^2(\mathcal{X}) \quad \int_{\mathcal{X} \times \mathcal{X}} K(\bar{x}, \hat{x}) f(\bar{x}) f(\hat{x}) d\bar{x} d\hat{x} \geq 0.$$

UNIVERSITÀ DEGLI STUDI DI BERGAMO

Kernels

In the finite dimensional case the above results can be simplified as follows.

Theorem

Let $K : \mathbb{R}^N \times \mathbb{R}^N \to \mathbb{R}$ *be a symmetric function. The function* $K(\cdot, \cdot)$ *is a* **Kernel function if and only** *if the the matrix* $K \in \mathbb{R}^{P \times P}$ *with*

$$K_{pq} = K(x^p, x^q)$$

is **positive semidefinite** *for each training set composed of* P *vectors.*

UNIVERSITÀ DEGLI STUDI DI BERGAMO

Kernels

The most common Kernels are:

- Linear Kernel: $K(\bar{x}, \hat{x}) = \bar{x}^\top \hat{x}$.
- Polynomial Kernel: $K(\bar{x}, \hat{x}) = (\bar{x}^\top \hat{x})^d$.
- Gaussian[1] Kernel (Radial Basis Function): $K(\bar{x}, \hat{x}) = e^{-\frac{||\bar{x} - \hat{x}||^2}{2\sigma^2}}$.
- Sigmoidal Kernel: $K(\bar{x}, \hat{x}) = \tanh(\rho_1 \bar{x}^\top \hat{x} + \rho_2)$.

[1] The Gaussian Kernel is an inner product in an infinite dimensional space.

Kernels

Moreover, new Kernels can be constructed based on existing Kernels via the following operations[2]:

- Scalar multiplication, where $\rho \geq 0$: $K(\bar{x}, \hat{x}) = \rho K_1(\bar{x}, \hat{x})$.
- Adding a positive constant, where $\rho \geq 0$: $K(\bar{x}, \hat{x}) = K_1(\bar{x}, \hat{x}) + \rho$.
- Linear combination with non-negative weights: $K(\bar{x}, \hat{x}) = \sum_{j=1}^{J} \rho_j K_j(\bar{x}, \hat{x})$, where $\rho_j \geq 0$ for $j = 1, \ldots, J$.
- Product of two Kernels: $K(\bar{x}, \hat{x}) = K_1(\bar{x}, \hat{x}) \cdot K_2(\bar{x}, \hat{x})$.
- Polynomial functions of a Kernel: $K(\bar{x}, \hat{x}) = P\big(K_1(\bar{x}, \hat{x})\big)$, where $P(\cdot)$ is a polynomial with non-negative coefficients.
- Exponential function of a Kernel: $K(\bar{x}, \hat{x}) = \exp\big(K_1(\bar{x}, \hat{x})\big)$.

UNIVERSITÀ DEGLI STUDI DI BERGAMO

[2] Here $K_1(\cdot, \cdot), K_2(\cdot, \cdot), \ldots, K_J(\cdot, \cdot)$ are valid Kernels.